

ICT702 Data Visualisation

Data Dashboards

Module 8



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 2

Data Dashboard Purpose

Data dashboard **purpose**: to support an organization's operations, decision-making, and strategic planning.

Possible Data dashboard **objectives**:

- tracking KPIs and monitoring processes
- assessing attainment of goals and objectives
- developing/enhancing insight
- sharing information
- measuring performance
- forecasting and data exploration

Data Dashboard Users' Needs

A data dashboard should be designed to assist a particular user or group of users with specific tasks associated with the management of an organization.

- The dashboard design team must understand:
 - the needs of the dashboard's end users
 - how addressing these needs supports the organization's objectives
- This knowledge will help the dashboard developer determine:
 - what information the dashboard should include
 - the most effective manner for presenting it

Data Dashboard Display

Considerations on how the information is displayed on a dashboard should include:

- selection of appropriate types of charts
- effective use of preattentive attributes, Gestalt principles, and colour
- a practical layout that enables end-users to quickly find the information they need and relate information from various charts in the dashboard
- strategies for avoiding overcrowding and unnecessary complexity such as organizing information into subsets and using interactive tools
- the device used to access the dashboard, ambient lighting, size, and resolution of the display

Data Dashboard Context

The context for the information included in the data dashboard may be provided by:

- showing how a KPI varies over time
- comparing the value of a KPI to an organizational goal
- comparing the value of a KPI:
 - internally across divisions, departments, or geographies of an organization
 - externally across customers
 - externally across competitors or organizations in the same industry

Data Dashboard Engineering

The dashboard design team must understand how:

- to organize the charts and tables on the dashboard to facilitate users' analyses and limit eye travel
- to maintain and assess the data dashboard effectiveness
- the organization's objectives may shift in the future. Specifically:
 - will the data dashboard need to reflect these shifts?
 - what new KPIs are likely to become important to the organization in the future?
 - what may be the source or format for the future incorporation of new data into the dashboard?

Data Dashboard Organization

- Store the dashboard data source in a separate worksheet.
- The dashboard should not draw information directly from the source data worksheet.
- Extract the data needed to create a chart or table from the source data worksheet and store it in a separate worksheet.
- The dashboard's users should not be allowed to make permanent changes to the dashboard.

Data Dashboard Testing

The dashboard design team must test its work extensively at each step to minimize a data dashboard's errors, miscommunications, and misunderstandings.

Failure to do so may result in:

- the creation of a faulty final product, leading to poor decisions and missed opportunities
- time-consuming and costly revisions
- damaging the credibility of the data dashboard design team

End of Part 2

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